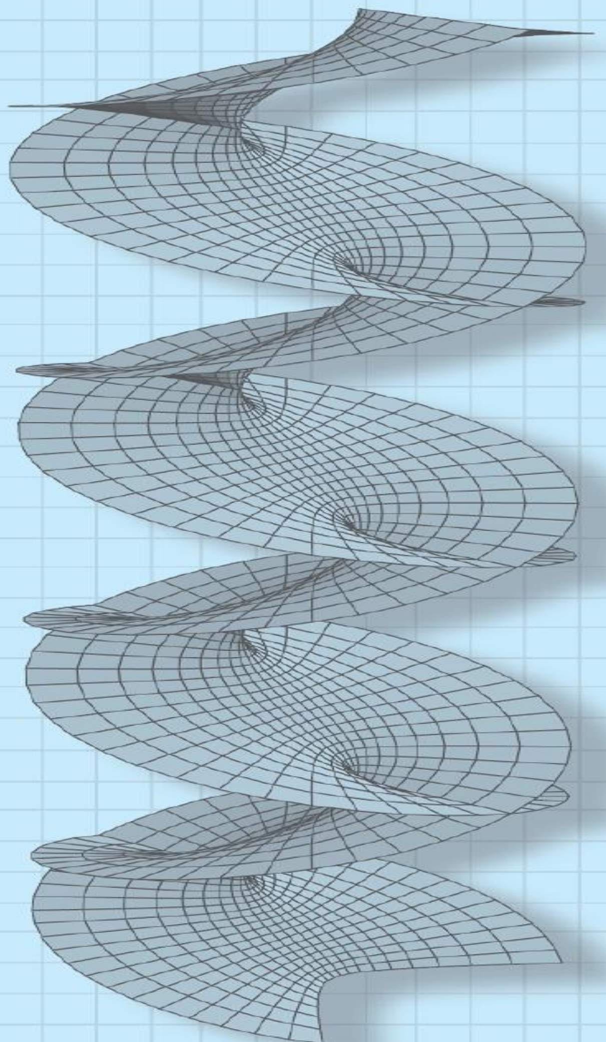

A First Course in

Complex Analysis

with Applications

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[redacted] Some Consequences of the Residue Theorem

In this section we shall see how residue theory can be used to evaluate *real* integrals of the forms

$$\int_0^{2\pi} F(\cos\theta, \sin\theta)\, d\theta, \tag{1}$$

$$\int_{-\infty}^{\infty} f(x)\, dx, \tag{2}$$

$$\int_{-\infty}^{\infty} f(x)\cos\alpha x\, dx \quad \text{and} \quad \int_{-\infty}^{\infty} f(x)\sin\alpha x\, dx, \tag{3}$$

where F in (1) and f in (2) and (3) are rational functions. For the rational function $f(x) = p(x)/q(x)$ in (2) and (3), we will assume that the polynomials p and q have no common factors.

In addition to evaluating the three integrals just given, we shall demonstrate how to use residues to evaluate real improper integrals that require integration along a branch cut.

The discussion ends with the relationship between the residue theory and the zeros of an analytic function and a consideration of how residues can, in certain cases, be used to find the sum of an infinite series.

Evaluation of Real Trigonometric Integrals

Integrals of the Form $\int_0^{2\pi} F(\cos\theta, \sin\theta)d\theta$

The basic idea here is to convert a real trigonometric integral of form (1) into a complex integral, where the contour C is the unit circle $|z| = 1$ centered at the origin.

To do this we begin with to parametrize this contour by $z = e^{i\theta}$, $0 \le \theta \le 2\pi$. We can then write

$$dz = ie^{i\theta}d\theta, \quad \cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

Since $dz = ie^{i\theta}d\theta = iz\,d\theta$ and $z^{-1} = 1/z = e^{-i\theta}$, these three quantities are equivalent to

$$d\theta = \frac{dz}{iz}, \quad \cos\theta = \frac{1}{2}(z + z^{-1}), \quad \sin\theta = \frac{1}{2i}(z - z^{-1}). \tag{4}$$

The conversion of the integral in (1) into a contour integral is accomplished by replacing, in turn, $d\theta$, $\cos\theta$, and $\sin\theta$ by the expressions in (4):

$$\oint_C F\left(\frac{1}{2}(z + z^{-1}), \frac{1}{2i}(z - z^{-1})\right)\frac{dz}{iz},$$

where C is the unit circle $|z| = 1$.

EXAMPLE 1 A Real Trigonometric Integral

Evaluate $\displaystyle\int_0^{2\pi} \frac{1}{(2+\cos\theta)^2}\,d\theta$.

Solution When we use the substitutions given in (4), the given trigonometric integral becomes the contour integral

$$\oint_C \frac{1}{\left(2+\frac{1}{2}(z+z^{-1})\right)^2}\frac{dz}{iz} = \oint_C \frac{1}{\left(2+\dfrac{z^2+1}{2z}\right)^2}\frac{dz}{iz}.$$

Carrying out the algebraic simplification of the integrand then yields

$$\frac{4}{i}\oint_C \frac{z}{(z^2+4z+1)^2}\,dz.$$

From the quadratic formula we can factor the polynomial z^2+4z+1 as $z^2+4z+1=(z-z_1)(z-z_2)$, where $z_1=-2-\sqrt{3}$ and $z_2=-2+\sqrt{3}$. Thus, the integrand can be written

$$\frac{z}{(z^2+4z+1)^2} = \frac{z}{(z-z_1)^2(z-z_2)^2}.$$

Because only z_2 is inside the unit circle C, we have

$$\oint_C \frac{z}{(z^2+4z+1)^2}\,dz = 2\pi i\,\mathrm{Res}(f(z),z_2).$$

To calculate the residue, we first note that z_2 is a pole of order 2 and so we use (2)

$$\begin{aligned}\operatorname{Res}(f(z), z_2) &= \lim_{z \to z_2} \frac{d}{dz}(z - z_2)^2 f(z) = \lim_{z \to z_2} \frac{d}{dz} \frac{z}{(z - z_1)^2} \\ &= \lim_{z \to z_2} \frac{-z - z_1}{(z - z_1)^3} = \frac{1}{6\sqrt{3}}.\end{aligned}$$

Hence, $$\frac{4}{i} \oint_C \frac{z}{(z^2 + 4z + 1)}\, dz = \frac{4}{i} \cdot 2\pi i \operatorname{Res}(f(z), z_1) = \frac{4}{i} \cdot 2\pi i \cdot \frac{1}{6\sqrt{3}}$$

and, finally, $$\int_0^{2\pi} \frac{1}{(2 + \cos\theta)^2}\, d\theta = \frac{4\pi}{3\sqrt{3}}.$$

Show by the present method that $\displaystyle\int_0^{2\pi} \frac{d\theta}{\sqrt{2} - \cos\theta} = 2\pi.$

Solution. We use $\cos\theta = \frac{1}{2}(z + 1/z)$ and $d\theta = dz/iz$. Then the integral becomes

$$\oint_C \frac{dz/iz}{\sqrt{2} - \frac{1}{2}\left(z + \frac{1}{z}\right)} = \oint_C \frac{dz}{-\frac{i}{2}(z^2 - 2\sqrt{2}z + 1)}$$

$$= -\frac{2}{i} \oint_C \frac{dz}{(z - \sqrt{2} - 1)(z - \sqrt{2} + 1)}.$$

We see that the integrand has a simple pole at $z_1 = \sqrt{2} + 1$ outside the unit circle C, so that it is of no interest here, and another simple pole at $z_2 = \sqrt{2} - 1$ (where $z - \sqrt{2} + 1 = 0$) inside C with residue

$$\operatorname*{Res}_{z=z_2} \frac{1}{(z - \sqrt{2} - 1)(z - \sqrt{2} + 1)} = \left[\frac{1}{z - \sqrt{2} - 1}\right]_{z=\sqrt{2}-1}$$

$$= -\frac{1}{2}.$$

Answer: $2\pi i(-2/i)(-\frac{1}{2}) = 2\pi$. (Here $-2/i$ is the factor in front of the last integral.)

Integrals of the Form $\int_{-\infty}^{\infty} f(x)\,dx$ Suppose $y = f(x)$ is a real function that is defined and continuous on the interval $[0, \infty)$. In elementary calculus the improper integral $I_1 = \int_0^\infty f(x)\,dx$ is defined as the limit

$$I_1 = \int_0^\infty f(x)\,dx = \lim_{R\to\infty} \int_0^R f(x)\,dx. \tag{5}$$

If the limit exists, the integral I_1 is said to be **convergent**; otherwise, it is **divergent**. The improper integral $I_2 = \int_{-\infty}^0 f(x)\,dx$ is defined similarly:

$$I_2 = \int_{-\infty}^0 f(x)\,dx = \lim_{R\to\infty} \int_{-R}^0 f(x)\,dx. \tag{6}$$

Finally, if f is continuous on $(-\infty, \infty)$, then $\int_{-\infty}^{\infty} f(x)\,dx$ is defined to be

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^0 f(x)\,dx + \int_0^\infty f(x)\,dx = I_1 + I_2, \tag{7}$$

provided *both* integrals I_1 and I_2 are convergent. If either one, I_1 or I_2, is divergent, then $\int_{-\infty}^{\infty} f(x)\,dx$ is divergent. It is important to remember that the right-hand side of (7) is *not* the same as

$$\lim_{R\to\infty} \left[\int_{-R}^0 f(x)\,dx + \int_0^R f(x)\,dx \right] = \lim_{R\to\infty} \int_{-R}^R f(x)\,dx. \tag{8}$$

For the integral $\int_{-\infty}^{\infty} f(x)\,dx$ to be convergent, the limits (5) and (6) must exist independently of one another. *But*, in the event that we know (a priori) that an improper integral $\int_{-\infty}^{\infty} f(x)\,dx$ converges, we can then evaluate it by means of the single limiting process given in (8):

$$\int_{-\infty}^{\infty} f(x)\,dx = \lim_{R\to\infty} \int_{-R}^{R} f(x)\,dx. \tag{9}$$

On the other hand, the symmetric limit in (9) may exist even though the improper integral $\int_{-\infty}^{\infty} f(x)\,dx$ is divergent. For example, the integral $\int_{-\infty}^{\infty} x\,dx$ is divergent since $\lim_{R\to\infty} \int_{0}^{R} x\,dx = \lim_{R\to\infty} \frac{1}{2}R^2 = \infty$. However, (9) gives

$$\lim_{R\to\infty} \int_{-R}^{R} x\,dx = \lim_{R\to\infty} \frac{1}{2}[R^2 - (-R)^2] = 0. \tag{10}$$

The limit in (9), if it exists, is called the **Cauchy principal value** (P.V.) of the integral and is written

$$\text{P.V.} \int_{-\infty}^{\infty} f(x)\,dx = \lim_{R\to\infty} \int_{-R}^{R} f(x)\,dx. \tag{11}$$

In (10) we have shown that P.V. $\int_{-\infty}^{\infty} x\,dx = 0$. To summarize:

Cauchy Principal Value

When an integral of form (2) converges, its Cauchy principal value is the same as the value of the integral. If the integral diverges, it may still possess a Cauchy principal value (11).

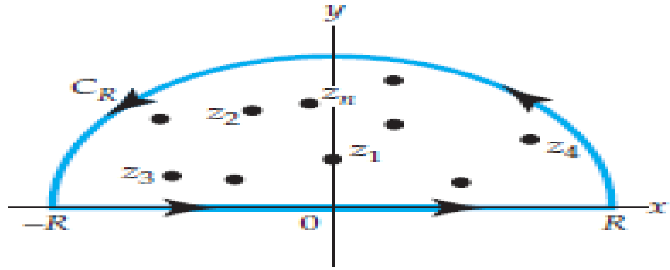
One final point about the Cauchy principal value: Suppose $f(x)$ is continuous on $(-\infty, \infty)$ and is an *even* function, that is, $f(-x) = f(x)$. Then its graph is symmetric with respect to the y-axis and as a consequence

$$\int_{-R}^{0} f(x)\,dx = \int_{0}^{R} f(x)\,dx \tag{12}$$

and
$$\int_{-R}^{R} f(x)\,dx = \int_{-R}^{0} f(x)\,dx + \int_{0}^{R} f(x)\,dx \;=\; 2\int_{0}^{R} f(x)\,dx. \tag{13}$$

From (12) and (13) we conclude that *if* the Cauchy principal value (11) exists, then both $\int_0^\infty f(x)\,dx$ and $\int_{-\infty}^{\infty} f(x)\,dx$ converge. The values of the integrals are

$$\int_{0}^{\infty} f(x)\,dx = \frac{1}{2}\,\text{P.V.}\int_{-\infty}^{\infty} f(x)\,dx \quad \text{and} \quad \int_{-\infty}^{\infty} f(x)\,dx = \text{P.V.}\int_{-\infty}^{\infty} f(x)\,dx.$$



To evaluate an integral $\int_{-\infty}^{\infty} f(x)\,dx$, where the rational function $f(x) = p(x)/q(x)$ is continuous on $(-\infty,\ \infty)$, by residue theory we replace x by the complex variable z and integrate the complex function f over a closed contour C that consists of the interval $[-R,\ R]$ on the real axis and a semicircle C_R of radius large enough to enclose all the poles of $f(z) = p(z)/q(z)$ in the upper half-plane $\mathrm{Im}(z) > 0$. we have

$$\oint_C f(z)\,dz = \int_{C_R} f(z)\,dz + \int_{-R}^{R} f(x)\,dx = 2\pi i \sum_{k=1}^{n} \mathrm{Res}(f(z), z_k),$$

where z_k, $k = 1, 2, \ldots, n$ denotes poles in the upper half-plane. If we can show that the integral $\int_{C_R} f(z)\,dz \to 0$ as $R \to \infty$, then we have

$$\text{P.V.} \int_{-\infty}^{\infty} f(x)\,dx = \lim_{R\to\infty} \int_{-R}^{R} f(x)\,dx = 2\pi i \sum_{k=1}^{n} \mathrm{Res}(f(z), z_k). \tag{14}$$

EXAMPLE 2 Cauchy P.V. of an Improper Integral

Evaluate the Cauchy principal value of $\displaystyle\int_{-\infty}^{\infty} \frac{1}{(x^2+1)(x^2+9)}\,dx$.

Solution Let $f(z) = 1/(z^2+1)(z^2+9)$. Since

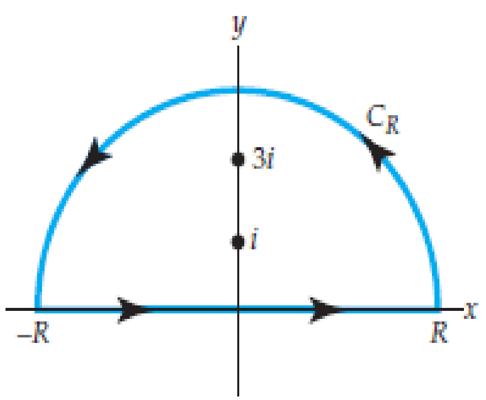
$$(z^2+1)(z^2+9) = (z-i)(z+i)(z-3i)(z+3i),$$

we take C be the closed contour consisting of the interval $[-R,\ R]$ on the x-axis and the semicircle C_R of radius $R > 3$. As seen from Figure

$$\oint_C \frac{1}{(z^2+1)(z^2+9)}\,dz = \int_{-R}^{R} \frac{1}{(x^2+1)(x^2+9)}\,dx + \int_{C_R} \frac{1}{(z^2+1)(z^2+9)}\,dz$$

$$= I_1 + I_2$$

and
$$I_1 + I_2 = 2\pi i\,[\mathrm{Res}(f(z), i) + \mathrm{Res}(f(z), 3i)].$$



At the simple poles $z = i$ and $z = 3i$ we find, respectively,

$$\operatorname{Res}(f(z),\ i) = \frac{1}{16i} \quad \text{and} \quad \operatorname{Res}(f(z),\ 3i) = -\frac{1}{48i},$$

so that
$$I_1 + I_2 = 2\pi i\left[\frac{1}{16i} + \left(-\frac{1}{48i}\right)\right] = \frac{\pi}{12}. \tag{15}$$

We now want to let $R \to \infty$ in (15). Before doing this, we use the inequality (10) of Section 1.2 to note that on the contour C_R,

$$\left|(z^2+1)(z^2+9)\right| = |z^2+1| \cdot |z^2+9| \geq \left|\,|z^2|\, - 1\right| \cdot \left|\,|z^2|\, - 9\right| = (R^2-1)(R^2-1)(R^2-9).$$

Since the length L of the semicircle is πR, it follows from the ML-inequality, that

$$|I_2| = \left|\int_{C_R} \frac{1}{(z^2+1)(z^2+9)}\,dz\right| \leq \frac{\pi R}{(R^2-1)(R^2-9)}.$$

This last result shows that $|I_2| \to 0$ as $R \to \infty$, and so we conclude that $\lim_{R\to\infty} I_2 = 0$. It follows from (15) that $\lim_{R\to\infty} I_1 = \pi/12$; in other words,

$$\lim_{R\to\infty}\int_{-R}^{R} \frac{1}{(x^2+1)(x^2+9)}\,dx = \frac{\pi}{12} \quad \text{or} \quad \text{P.V.}\int_{-\infty}^{\infty} \frac{1}{(x^2+1)(x^2+9)}\,dx = \frac{\pi}{12}.$$

Theorem **Behavior of Integral as $R \to \infty$**

Suppose $f(z) = \dfrac{p(z)}{q(z)}$ is a rational function, where the degree of $p(z)$ is n and the degree of $q(z)$ is $m \geq n+2$. If C_R is a semicircular contour $z = Re^{i\theta}$, $0 \leq \theta \leq \pi$, then $\int_{C_R} f(z)\, dz \to 0$ as $R \to \infty$.

In other words, the integral along C_R approaches zero as $R \to \infty$ when the denominator of f is of a power at least 2 more than its numerator. The proof of this fact follows in the same manner as in Example 2. Notice in that example, the conditions stipulated in Theorem are satisfied, since degree of $p(z) = 1$ is 0 and the degree of $q(z) = (z^2+1)(z^2+9)$ is 4.

EXAMPLE 3 Cauchy P.V. of an Improper Integral

Evaluate the Cauchy principal value of $\int_{-\infty}^{\infty} \frac{1}{x^4+1}\, dx$.

Solution By inspection of the integrand we see that the conditions given in Theorem are satisfied. Moreover, we know that $f(z) = 1/(z^4+1)$ has simple poles in the upper half-plane at $z_1 = e^{\pi i/4}$ and $z_2 = e^{3\pi i/4}$. We also saw in that example that the residues at these poles are

$$\operatorname{Res}(f(z), z_1) = -\frac{1}{4\sqrt{2}} - \frac{1}{4\sqrt{2}} i \quad \text{and} \quad \operatorname{Res}(f(z), z_2) = \frac{1}{4\sqrt{2}} - \frac{1}{4\sqrt{2}} i \,.$$

Thus, by (14),

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{1}{x^4+1}\, dx = 2\pi i \left[\operatorname{Res}(f(z), z_1) + \operatorname{Res}(f(z), z_2)\right] = \frac{\pi}{\sqrt{2}}.$$

Since the integrand is an even function, the original integral converges to $\pi/\sqrt{2}$.

Integrals of the Form $\int_{-\infty}^{\infty} f(x)\cos\alpha\, x\, dx$ and $\int_{-\infty}^{\infty} f(x)\sin\alpha\, x\, dx$

Because improper integrals of the form $\int_{-\infty}^{\infty} f(x)\sin\alpha\, x\, dx$ are encountered in applications of Fourier analysis, they often are referred to as **Fourier integrals.** Fourier integrals appear as the real and imaginary parts in the improper integral $\int_{-\infty}^{\infty} f(x)e^{i\alpha x}\, dx$.[‡] In view

of Euler's formula $e^{i\alpha x} = \cos\alpha x + i\sin\alpha x$, where α is a positive real number, we can write

$$\int_{-\infty}^{\infty} f(x)\, e^{i\alpha x} dx = \int_{-\infty}^{\infty} f(x)\cos\alpha x\, dx + i\int_{-\infty}^{\infty} f(x)\sin\alpha x\, dx \tag{16}$$

whenever both integrals on the right-hand side converge. Suppose $f(x) = p(x)/q(x)$ is a rational function that is continuous on $(-\infty, \infty)$. Then both Fourier integrals in (10) can be evaluated at the same time by considering the complex integral $\int_C f(z)e^{i\alpha z} dz$, where $\alpha > 0$, and the contour C again consists of the interval $[-R, R]$ on the real axis and a semicircular contour C_R with radius large enough to enclose the poles of $f(z)$ in the upper-half plane.

Before proceeding, we give, without proof, sufficient conditions under which the contour integral along C_R approaches zero as $R \to \infty$.

Theorem Behavior of Integral as $R \to \infty$

Suppose $f(z) = \dfrac{p(z)}{q(z)}$ is a rational function, where the degree of $p(z)$ is n and the degree of $q(z)$ is $m \geq n+2$. If C_R is a semicircular contour $z = Re^{i\theta}$, $0 \leq \theta \leq \pi$, and $\alpha > 0$, then $\int_{C_R} f(z)\, e^{i\alpha z} dz \to 0$ as $R \to \infty$.

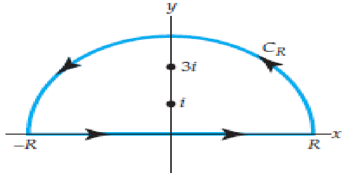
EXAMPLE 4 Using Symmetry

Evaluate the Cauchy principal value of $\displaystyle\int_0^{\infty} \frac{x \sin x}{x^2+9}\, dx$.

Solution First note that the limits of integration in the given integral are not from $-\infty$ to ∞ as required by the method just described. This can be remedied by observing that since the integrand is an even function of x (verify), we can write

$$\int_0^{\infty} \frac{x \sin x}{x^2+9}\, dx = \frac{1}{2}\int_{-\infty}^{\infty} \frac{x \sin x}{x^2+9}\, dx. \tag{17}$$

With $\alpha = 1$ we now form the contour integral



$$\oint_C \frac{z}{z^2+9}\, e^{iz}\, dz,$$

where C is the same contour shown in Figure !.

$$\int_{C_R} \frac{z}{z^2+9} e^{iz} dz + \int_{-R}^{R} \frac{x}{x^2+9} e^{ix}\, dx = 2\pi i\ \mathrm{Res}(f(z)e^{iz}, 3i),$$

where $f(z) = z/(z^2+9)$, and

$$\mathrm{Res}\Big(f(z)e^{iz},\ 3i\Big) = \left.\frac{ze^{iz}}{2z}\right|_{z=3i} = \frac{e^{-3}}{2}$$

last

Then, from Theorem we conclude $\int_{C_R} f(z)e^{iz}\,dz \to 0$ as $R \to \infty$, and so

$$\text{P.V.}\int_{-\infty}^{\infty} \frac{x}{x^2+9}\,e^{ix}\,dx = 2\pi i\left(\frac{e^{-3}}{2}\right) = \frac{\pi}{e^3}i.$$

But by (16),

$$\int_{-\infty}^{\infty} \frac{x}{x^2+9}\,e^{ix}\,dx = \int_{-\infty}^{\infty} \frac{x\cos x}{x^2+9}\,dx + i\int_{-\infty}^{\infty} \frac{x\sin x}{x^2+9}\,dx = \frac{\pi}{e^3}i.$$

Equating real and imaginary parts in the last line gives the bonus result

$$\text{P.V.}\int_{-\infty}^{\infty} \frac{x\cos x}{x^2+9}\,dx = 0 \qquad \text{along with} \qquad \text{P.V.}\int_{-\infty}^{\infty} \frac{x\sin x}{x^2+9}\,dx = \frac{\pi}{e^3}. \quad (18)$$

Finally, in view of the fact that the integrand is an even function, we obtain the value of the prescribed integral:

$$\int_{0}^{\infty} \frac{x\sin x}{x^2+9}\,dx = \frac{1}{2}\int_{-\infty}^{\infty} \frac{x\sin x}{x^2+9}\,dx = \frac{\pi}{2e^3}.$$

$$\int_{-\infty}^{\infty} f(x)e^{isx}\,dx = 2\pi i \sum \operatorname{Res} \ f(z)e^{isz}$$

$$\int_{-\infty}^{\infty} f(x)\cos sx\,dx = -2\pi \sum \operatorname{Im} \operatorname{Res} \ f(z)e^{isz} \ ,$$

$$\int_{-\infty}^{\infty} f(x)\sin sx\,dx = 2\pi \sum \operatorname{Re} \operatorname{Res} \ f(z)e^{isz} \ .$$

$$(s > 0)$$

Show that $\displaystyle\int_{-\infty}^{\infty} \frac{\cos sx}{k^2+x^2}\,dx = \frac{\pi}{k}e^{-ks}, \qquad \int_{-\infty}^{\infty} \frac{\sin sx}{k^2+x^2}\,dx = 0 \qquad (s>0, k>0).$

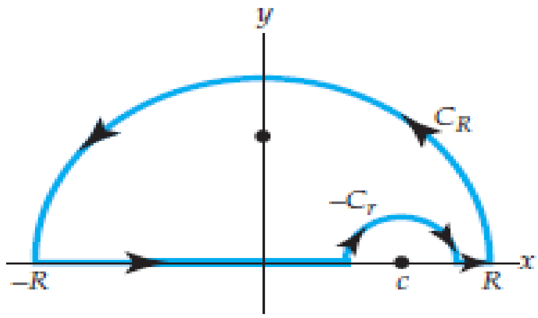
Solution. In fact, $e^{isz}/(k^2+z^2)$ has only one pole in the upper half-plane, namely, a simple pole at $z = ik$, we obtain

$$\operatorname*{Res}_{z=ik} \frac{e^{isz}}{k^2+z^2} = \left[\frac{e^{isz}}{2z}\right]_{z=ik} = \frac{e^{-ks}}{2ik}.$$

Thus

$$\int_{-\infty}^{\infty} \frac{e^{isx}}{k^2+x^2}\,dx = 2\pi i \frac{e^{-ks}}{2ik} = \frac{\pi}{k}e^{-ks}.$$

Since $e^{isx} = \cos sx + i \sin sx$, this yields the above results



Indented contour

Indented Contours The improper integrals of forms (2) and (3) that we have considered up to this point were continuous on the interval $(-\infty, \infty)$. In other words, the complex function $f(z) = p(z)/q(z)$ did not have poles on the real axis. In the situation where f has poles on the real axis, we must modify the procedure illustrated in Examples 2–4. For example, to evaluate $\int_{-\infty}^{\infty} f(x)\,dx$ by residues when $f(z)$ has a pole at $z = c$, where c is a real number, we use an **indented contour** as illustrated in Figure . The symbol C_r denotes a semicircular contour centered at $z = c$ and oriented in the *positive* direction. The next theorem is important to this discussion.

Theorem **Behavior of Integral as $r \to 0$**

Suppose f has a simple pole $z = c$ on the real axis. If C_r is the contour defined by $z = c + re^{i\theta}$, $0 \le \theta \le \pi$, then

$$\lim_{r\to 0}\int_{C_r} f(z)\,dz = \pi i\,\mathrm{Res}(f(z), c).$$

Proof Since f has a simple pole at $z = c$, its Laurent series is

$$f(z) = \frac{a_{-1}}{z-c} + g(z),$$

where $a_{-1} = \mathrm{Res}(f(z),\ c)$ and g is analytic at the point c. Using the Laurent series and the parametrization of C_r we have

$$\int_{C_r} f(z)\,dz = a_{-1}\int_0^{\pi} \frac{ire^{i\theta}}{re^{i\theta}}\,d\theta + ir\int_0^{\pi} g(c+re^{i\theta})\,e^{i\theta}d\theta = I_1 + I_2. \qquad (19)$$

First, we see that

$$I_1 = a_{-1} \int_0^{\pi} \frac{ire^{i\theta}}{re^{i\theta}}\, d\theta = a_{-1} \int_0^{\pi} i\, d\theta = \pi i a_{-1} = \pi i \operatorname{Res}(f(z), c).$$

Next, g is analytic at c, and so it is continuous at this point and bounded in a neighborhood of the point; that is, there exists an $M > 0$ for which $\left|g(c + re^{i\theta})\right| \leq M$. Hence,

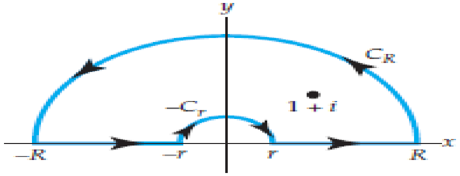
$$|I_2| = \left| ir \int_0^{\pi} g(c + re^{i\theta})\, d\theta \right| \leq r \int_0^{\pi} M\, d\theta = \pi r M.$$

It follows from this last inequality that $\lim_{r\to 0} |I_2| = 0$ and consequently $\lim_{r\to 0} I_2 = 0$. By taking the limit of (19) as $r \to 0$, the theorem is proved.

EXAMPLE 5 Using an Indented Contour

Evaluate the Cauchy principal value of $\displaystyle\int_{-\infty}^{\infty} \frac{\sin x}{x(x^2-2x+2)}\,dx$.

Solution Since the integral is of the type given in (3), we consider the contour integral



$$\oint_C \frac{e^{iz}}{z(z^2-2z+2)}\,dz.$$

The function $f(z) = 1/z(z^2-2z+2)$ has a pole at $z=0$ and at $z=1+i$ in the upper half-plane. The contour C, shown in Figure , is indented at the origin. Adopting an obvious condensed notation, we have

$$\oint_C = \int_{C_R} + \int_{-R}^{-r} + \int_{-C_r} + \int_r^R = 2\pi i\,\mathrm{Res}(f(z)e^{iz}, 1+i), \qquad (20)$$

where $\int_{-C_r} = -\int_{C_r}$. If we take the limits of (20) as $R\to\infty$ and as $r\to 0$, it follows from Theorems that

$$\text{P.V.}\int_{-\infty}^{\infty} \frac{e^{ix}}{x(x^2-2x+2)}\,dx - \pi i\,\mathrm{Res}(f(z)e^{iz},0) = 2\pi i\,\mathrm{Res}(f(z)e^{iz},1+i).$$

Now,

$$\mathrm{Res}(f(z)e^{iz},0) = \frac{1}{2} \quad \text{and} \quad \mathrm{Res}(f(z)e^{iz},1+i) = -\frac{e^{-1+i}}{4}(1+i).$$

Therefore,

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{e^{ix}}{x(x^2 - 2x + 2)}\, dx = \pi i \left(\frac{1}{2} \right) \; + \; 2\pi i \left(-\frac{e^{-1+i}}{4} (1+i) \right).$$

Using $e^{-1+i} = e^{-1}(\cos 1 + i \sin 1)$, simplifying, and then equating real and imaginary parts, we get from the last equality

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{\cos x}{x(x^2 - 2x + 2)}\, dx = \frac{\pi}{2} e^{-1}(\sin 1 + \cos 1)$$

and $$\text{P.V.} \int_{-\infty}^{\infty} \frac{\sin x}{x(x^2 - 2x + 2)}\, dx = \frac{\pi}{2}[1 + e^{-1}(\sin 1 - \cos 1)].$$

Simple Poles on the Real Axis

If $f(z)$ has a simple pole at $z = a$ on the real axis, then

$$\lim_{r\to 0} \int_{C_2} f(z)\, dz = \pi i \operatorname*{Res}_{z=a} f(z).$$

$$\text{pr. v.} \int_{-\infty}^{\infty} f(x)\, dx = 2\pi i \sum \operatorname{Res} f(z) + \pi i \sum \operatorname{Res} f(z)$$

where the first sum extends over all poles in the upper half-plane and the second over all poles on the real axis, the latter being simple by assumption.

Find the principal value

$$\text{pr. v.} \int_{-\infty}^{\infty} \frac{dx}{(x^2 - 3x + 2)(x^2 + 1)} .$$

Solution. Since

$$x^2 - 3x + 2 = (x - 1)(x - 2),$$

the integrand $f(x)$, considered for complex z, has simple poles at

$$z = 1, \qquad \operatorname*{Res}_{z=1} f(z) = \left[\frac{1}{(z - 2)(z^2 + 1)}\right]_{z=1} = -\frac{1}{2},$$

$$z = 2, \qquad \operatorname*{Res}_{z=2} f(z) = \left[\frac{1}{(z - 1)(z^2 + 1)}\right]_{z=2} = \frac{1}{5},$$

$$z = i, \qquad \operatorname*{Res}_{z=i} f(z) = \left[\frac{1}{(z^2 - 3z + 2)(z + i)}\right]_{z=i} = \frac{1}{6 + 2i} = \frac{3 - i}{20},$$

and at $z = -i$ in the lower half-plane, which is of no interest here. From we get the answer

$$\text{pr. v.} \int_{-\infty}^{\infty} \frac{dx}{(x^2 - 3x + 2)(x^2 + 1)} = 2\pi i\left(\frac{3 - i}{20}\right) + \pi i\left(-\frac{1}{2} + \frac{1}{5}\right) = \frac{\pi}{10} .$$